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Toward a Free-Response Paradigm of Decision Making in Spiking Neural Networks

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Spiking neural networks (SNNs) have attracted significant interest in the development of brain-inspired computing systems due to their energy efficiency and similarities to biological information processing. In contrast to continuous-valued artificial neural networks, which produce results in a single step, SNNs require multiple steps during inference to achieve a desired accuracy level, resulting in a burden in real-time response and energy efficiency. Inspired by the tradeoff between speed and accuracy in human and animal decision-making processes, which exhibit correlations among reaction times, task complexity, and decision confidence, an inquiry emerges regarding how an SNN model can benefit by implementing these attributes. Here, we introduce a theory of decision making in SNNs by untangling the interplay between signal and noise. Under this theory, we introduce a new learning objective that trains an SNN not only to make the correct decisions but also to shape its confidence. Numerical experiments demonstrate that SNNs trained in this way exhibit improved confidence expression, reduced trial-to-trial variability, and shorter latency to reach the desired accuracy. We then introduce a stopping policy that can stop inference in a way that further enhances the time efficiency of SNNs. The stopping time can serve as an indicator to whether a decision is correct, akin to the reaction time in animal behavior experiments. By integrating stochasticity into decision making, this study opens up new possibilities to explore the capabilities of SNNs and advance SNNs and their applications in complex decision-making scenarios where model performance is limited.

1 Introduction

Spiking neural networks (SNNs), regarded as the third generation of neural networks, emulate the processing mechanisms of the biological brain and are believed to offer the potential to perform more intricate cognitive functions (Maass, 1997; Roy et al., 2019). However, compared to continuous-valued artificial neural networks (ANNs), which have made significant advances in various fields (LeCun et al., 2015; Wang et al., 2023), it is still unclear how the potential of SNNs can be realized (Li, Deng et al., 2024; Neftci et al., 2019; Pfeiffer & Pfeil, 2018). Specifically, despite numerous successful studies on the training of SNNs (Bellec et al., 2020; Deng et al., 2022; Qi et al., 2023; Rueckauer et al., 2017; Shrestha & Orchard, 2018; Wu et al., 2018, 2022), a common challenge for these networks is the intrinsic trade-off between speed and accuracy (Deng et al., 2020; Dong et al., 2024), where higher performance often requires considerably longer simulation duration.

Rather than focusing solely on reducing the latency of SNN to achieve desired accuracy, our objective in this work is to integrate human cognitive decision making principles, which interpret latency as a measure of decision confidence. The core aspect of perceptual decision making involves

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the accumulation of evidence, which addresses uncertainties in the environment by integrating information over time (Churchland et al., 2008; Gold & Shadlen, 2007; Mazurek et al., 2003; Palmer et al., 2005). This integration process embodies the speed-accuracy trade-off (Bogacz et al., 2010; Gold & Shadlen, 2007): the probability of a correct decision improves with the gradual aggregation of evidence, yet this collection of information inherently demands time. The reaction time in humans and animals, which captures the duration from stimulus presentation to response, varies with task difficulty (Kepecs et al., 2008; Kiani et al., 2014) and is significantly shorter for correct decisions compared to incorrect ones (Deco et al., 2009; Kiani et al., 2014; Vickers & Packer, 1982), indicating its potential as a confidence measure in decision making.

Random drift-diffusion models (DDMs) are widely used to understand the fundamental aspects of these psychophysical processes (Bogacz et al., 2006; Gold & Shadlen, 2007; Ratcliff & McKoon, 2008; Ratcliff & Smith, 2004). Within the domain of DDMs, decision-making processes involve accumulating evidence, which is depicted as the integral of a stochastic process over time until a specific decision threshold is reached. This conceptual framework not only corresponds to experimental findings but also suggests that the brain employs probabilistic reasoning through sampling of neural activity (Haefner et al., 2016; Hoyer & Hyvärinen, 2002; Savin & Denève, 2014). Therefore, the decision that an agent makes is primarily influenced by the distribution from which the evidence is drawn and by the structure of its mean and covariance in relation to the decision boundary (Averbeck et al., 2006; Azeredo da Silveira & Rieke, 2021; Panzeri et al., 2022).

In this study, we explore the SNN inference processes to establish a theoretical link between decision making and decision confidence, in line with the principle used in DDMs. This theory explains how the mean and covariance of evidence distributions, along with their relation to the decision boundary, affect an SNN's decision and the time needed for a stable outcome. Moreover, it underscores a shift from the conventional interrogation paradigm of inference in SNNs, where decisions are made with a fixed latency regardless of input variations, to the free-response paradigm, where SNNs dynamically stop the inference process based on the confidence level of the current decision. In particular, the free-response paradigm is the fastest method to make a decision while maintaining a given level of accuracy in DDM (Bogacz et al., 2006).

According to this theory, we present a fidelity-entropy loss function aimed at training SNNs to not only make correct decisions but also enhance their confidence expression based on the decision outcomes. Through numerical experiments on tasks with varying difficulty levels, we demonstrate that this loss function enhances the model's capability to express confidence and reduces inference latency by optimizing the covariance of evidence distributions. Furthermore, our theory proposes a straightforward and effective stopping policy that enhances the speed of inference by

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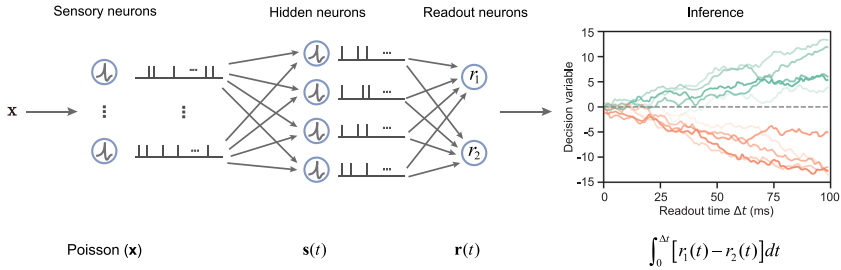


Figure 1: The principle of decision making in SNN with drift-diffusion model. Sensory neurons encode stimulus $\mathbf{x}$ into corresponding spike trains via a Poisson encoding scheme (Poisson($\mathbf{x}$)), which serves as input to a downstream network. After layers of processing, a linear readout then transforms the spiking activity of the last hidden layer $\mathbf{s}(t)$ into evidence $\mathbf{r}(t)$ for decision making. The decision variable integrates and compares evidence over time ($\int_0^{\Delta t}[r_1(t) - r_2(t)dt]$), emulating evidence accumulation in the drift-diffusion model.

flexible termination, with the stopping time reflecting the decision's confidence. This adaptive stopping policy operates independent of the model training process, and its effectiveness is demonstrated across multiple SNN training frameworks. This reveals a fundamental characteristic of decision making in SNNs, offering a novel perspective on their accuracy-latency trade-off properties.

2 Results

2.1 Theoretical Framework for Decision Making in Spiking Neural Networks. We begin with an overview of the inference process in SNNs, as illustrated in Figure 1. The external stimulus $\mathbf{x}$ (e.g., an image) is initially recoded by the spike activity of the sensory neurons through a Poisson rate coding in which sensory neurons emit independent Poisson spike trains with mean firing rate proportional to the intensity of each image pixel. These spike trains are then used as input to the SNN in a feedforward fashion. Let the neural activity in the last hidden layer of the network at time t be $\mathbf{s}(t)$. The decision-making mechanism employs a linear decoder W to continuously map $\mathbf{s}(t)$ into a readout $\mathbf{r}(t)$:

$$\mathbf{r}(t) = W\mathbf{s}(t). \tag{2.1}$$

The instantaneous readout value $\mathbf{r}(t)$ is inherently random due to the input noise originating from Poisson spike trains and internal network variations, such as randomly distributed initial membrane potentials. Impreciseness can also arise from limited time resolutions and restricted

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precisions in synaptic weights of model implementation on specific hardwares. Consequently, instantaneous readout $\mathbf{r}(t)$ can be modeled as sampling from a conditional distribution $p(\mathbf{r}|\mathbf{x}, \theta)$ (Savin & Denève, 2014; Zhang et al., 2023), where θ represents the model parameters. In this context, the readout of SNNs is equivalent to the evidence used for decision making. Hence, the terms *evidence* and *readout* can be used interchangeably to refer to models' output. Similarly, the terms *prediction* and *decision* can be interchangeably used to describe the model's choice.

Due to the inherent randomness of instantaneous readout $\mathbf{r}(t)$, a reliable inference requires accumulating the readout over an appropriate time. For this purpose, the cumulative readout $\mathbf{d}$ is used, which is defined as

$$\mathbf{d} = \int_0^{\Delta t} \mathbf{r}(t)dt, \tag{2.2}$$

where Δt is the duration of the readout. Assuming that each $\mathbf{r}(t)$ is drawn from a gaussian distribution with mean μ and covariance $\mathbf{\Sigma}$ independently, the accumulated evidence is $\mathbf{d} \sim \mathcal{N}(\mu\Delta t, \mathbf{\Sigma}\Delta t)$. Hence, the coefficient of variation (CV) for entry i in $\mathbf{d}$ is $\frac{\sigma_i}{\mu_i\sqrt{\Delta t}}$. Namely, as the amount of evidence accumulates over time, the noise in the readout progressively decreases, resulting in more reliable decision making.

If we model the probability of choosing category i by the softmax function,

$$P(i|\mathbf{d}) = \frac{\exp d_i}{\sum_j \exp d_j}, \tag{2.3}$$

then the log-odds ratio between category i and j is

$$\Lambda = \ln \frac{P(i|\mathbf{d})}{P(j|\mathbf{d})} = d_i - d_j. \tag{2.4}$$

Consequently, a greater difference between d_i and d_j suggests a higher probability of selecting the category i over j. Moreover, when the readout time Δt is long enough, the log-odds ratio approaches $(\mu_i - \mu_j)\Delta t$, where the sign is determined by the difference between μ_i and μ_j. This guarantees that the model will choose the category that has the highest μ with probability equal to one. However, an indefinitely long readout time is neither desirable nor necessary due to the importance of quick responses, especially when the decision remains the same and will not change in the future. Therefore, decision making in SNNs aims to answer two main questions: Which category should be selected as the most suitable choice, given the evidence available so far? And is the chosen option truly the best if we collect more evidence?

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The solution to the first question, determining the best decision, typically involves selecting the index of the entry with the greatest value in $\mathbf{d}$ to signify the choice it will make. Consider a binary decision-making task in which the decision variable (DV) is defined as

$$\mathrm{DV} = d_1 - d_2 = \int_0^{\Delta t} [r_1(t) - r_2(t)]\, dt, \tag{2.5}$$

where d_1, d_2 are two corresponding entries in $\mathbf{d}$. In this task, the model's choice is determined by whether the DV is positive or negative. Given the input noise, the sign of the DV trajectories over time (see Figure 1, right-most panel) may initially fluctuate but will stabilize after a long enough readout time, ensuring that the model will ultimately choose the option with the highest readout mean. However, a sufficiently long time for decision making is often infeasible in practice. Consequently, the second question arises that we need to assess the confidence level in the current decision, specifically if there is a risk of overturning the decision if allowing more time for additional evidence. As the standard deviation of the DV's magnitude increases more slowly than its mean, there is a moment when the DV's sign becomes stable. This moment marks the convergence time of the inference process, where a shorter convergence time implies a faster commitment to the decision in SNNs. Thus, the second inquiry can be reframed by evaluating the probability that an SNN can accurately identify the category whose mean readout is the highest.

Let the categories i and j have the highest and second-highest mean readout values; the probability that the network will make the best decision (selecting i) is equivalent to computing the probability $P(d_i - d_j > 0)$, which signifies the confidence of the model in its current decision. Assuming that instantaneous readouts $\mathbf{r}$ are sampled from a gaussian distribution with a mean μ and a covariance $\mathbf{\Sigma}$, we can derive an analytical formula for this probability as

$$P(d_i - d_j > 0) = \frac{1}{2}\mathrm{erfc}\left(-\frac{(\mu_i - \mu_j)\sqrt{\Delta t}}{\sqrt{2(\sigma_i^2 + \sigma_j^2 - 2\sigma_i\sigma_j\rho_{ij})}}\right), \tag{2.6}$$

where $(\mu_i, \mu_j, \sigma_i, \sigma_j)$ represent the mean and standard deviation of the respective categories, ρ_{ij} is the correlation coefficient between categories i and j, and erfc denotes the complementary error function. The key determinant influencing the decision is the quantity $\mu_i - \mu_j$, as this is the only factor dictating the direction of the confidence changes. Therefore, the confidence of the model in category i will increase only when μ_i is the highest mean among all categories as the readout time Δt increases. This criterion ensures that given enough readout time, the system will consistently prefer

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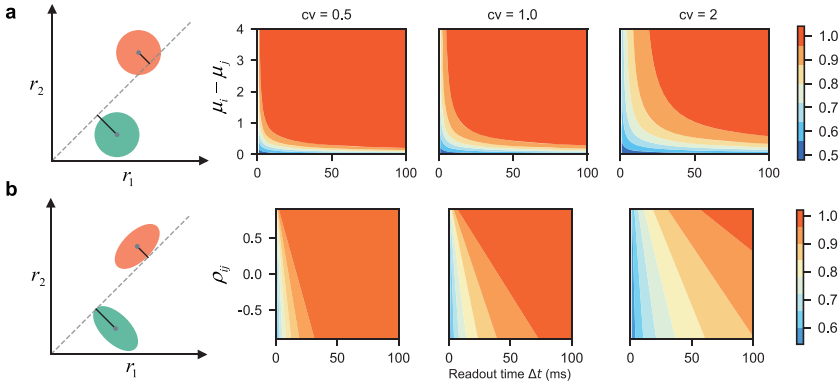


Figure 2: Temporal dynamics of model confidence during inference given varying configurations of readout distributions. (a) Assuming option i is the best ($\mu_i > \mu_j$, i can be either 1 or 2 and $i \neq j$), we investigate the evolution of the model's confidence $P(d_i - d_j > 0)$ over readout time Δt for varying values of μ_i and σ_i, adjusted by the coefficient of variation (CV), with fixed $\mu_j = \sigma_j = 1$ and $\rho_{ij} = 0$. (b) The impact of correlation between options i and j on decision confidence is analyzed by maintaining a fixed mean difference and varying CV. Changes in confidence $P(d_i - d_j > 0)$ under different scenarios in panels a and b are depicted using color maps.

the option with the highest mean value. Furthermore, a greater difference in means can accelerate the increase of $P(d_i - d_j > 0)$, resulting in a shorter readout time, as illustrated in Figure 2a. In a scenario with $\mu_i - \mu_j$ fixed, a higher variance can hinder this progression, as illustrated in the same graphical representations where ρ_{ij} is set at zero. Moreover, this impact can be influenced by the structure of covariance (see Figure 2b). When entries i and j exhibit a positive correlation, they lower the denominator in equation 2.6, thereby accelerating the growth of confidence. In contrast, negative correlations will result in the opposite effect, decelerating the confidence increase.

From the preceding discussion, we can distill two training objectives for SNNs. The primary goal is to optimize a model's ability to make correct decisions by guaranteeing that the correct category can have the highest readout value. The secondary goal is to optimize models' self-assessment ability during decision making, so that their confidence should be higher for correct predictions than incorrect predictions. Although the computation of model confidence is straightforward in binary decision-making tasks, it becomes challenging when dealing with multiclass scenarios. Specifically, confidence involves calculating the probability $q_i = \int p(\mathbf{d}|\mathbf{x}, \theta)\mathbf{1}_i(\mathbf{d})d\mathbf{d}$, where the indicator function $\mathbf{1}_i(\mathbf{d})$ equals one if d_i is the maximum among all entries and zero otherwise. Given the absence of

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analytic solutions for this probability and the significant computational burden of sampling-based estimations in high-dimensional spaces, we propose a more practical method by comparing the confidence levels between the entry with the highest value in the readout mean vector μ and the other entries pairwise. The expression can be simplified because, as detailed in equation 2.6, the mean vector μ becomes the dominant factor when Δt is large. That is, the influence of the denominator in equation 2.6 will decrease after an extended period of evidence collection. If the probability $P(d_i - d_j > 0)$ between the top two values in μ is sufficiently high, it suggests that $P(d_i - d_k > 0)$ is high for any $k \neq i$. This understanding leads to the formulation of a novel loss function for SNN optimization,

$$L(\theta) = \underbrace{\left(-\mu_{\text{label}} + \log \sum_k \exp \mu_k\right)}_{\text{cross-entropy loss}} \tag{2.7}$$

$$+ \underbrace{f(\text{label}, i)\left(-\sum_{k \neq i} w_k \left(P_{ik} \log P_{ik} + (1 - P_{ik}) \log(1 - P_{ik})\right)\right)}_{\text{fidelity-entropy loss}}, \tag{2.8}$$

where μ_{label} represents the mean readout of the target entry, P_{ik} denotes the confidence measurement, equation 2.6, between entry i, which has the highest readout mean value, and other entries, with w_k being nonnegative weights summing up to one. Here $f(\text{label}, i)$ is an indicator function that aligns optimization with the prediction results, producing 1 for correct predictions ($i = \text{label}$) and -1 for incorrect predictions ($i \neq \text{label}$). The cross-entropy component is focused on enhancing the mean of target entry, in line with conventional training methods for ANN and SNN. The fidelity entropy component is intended to calibrate the prediction confidence of a model based on whether its prediction is correct or incorrect. In this context, Δt functions as a hyperparameter that balances the optimization objectives between the numerator and the denominator of equation 2.6. Setting appropriate weight coefficients w_j can be challenging and can influence the effectiveness of model training. One guideline is that w_j should be higher for pairs that result in greater fidelity-entropy loss. In the main results presented here, we assign a set of default weight coefficients that works well. An analysis of the effect of different weight configurations on learning and inference is included in appendix B.

From the viewpoint of inference, if we adhere to the same principle in DDMs, where an SNN makes its prediction once its confidence in option i exceeds a threshold Θ such that $P(d_i - d_j > 0) > \Theta$, for all $i \neq j$, then the minimal readout time Δt required to reach a decision can be derived from equation 2.6 as

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$$\Delta t > (\text{erfc}^{-1}(2\Theta))^2 \frac{2(\sigma_i^2 + \sigma_j^2 - 2\sigma_i\sigma_j\rho_{ij})}{(\mu_i - \mu_j)^2}. \tag{2.9}$$

Since the readout mean and covariance are influenced by the input $\mathbf{x}$, the equation suggests a paradigm shift from an interrogation paradigm, in which the model makes decisions with a fixed readout time window regardless of input variations, to a free-response paradigm that uses confidence levels to determine when to make a decision, aligning with the principles seen in DDMs (Bogacz et al., 2006; Gold & Shadlen, 2007; Ratcliff & McKoon, 2008).

To validate our hypothesis, we use the moment neural network (MNN) as a substitute for SNN training (Feng et al., 2006; Lu et al., 2010; Qi et al., 2023). The MNN is chosen for its mathematical rigor and its ability to accurately capture the statistical properties and nonlinear dynamics inherent in the correlated fluctuations observed in spiking neurons. This ability enables direct optimization of the readout distribution, a crucial component of our methodology. A brief introduction to MNN is provided in section 4. In the following, we first examine the confidence expression ability of MNNs trained with the proposed fidelity-entropy loss in contrast to those trained without it. Next, we experimentally validate SNNs built with trained MNN parameters. This is intended to underscore the effectiveness of our theoretical framework. Finally, we examine various stopping policies to further enhance energy efficiency. These policies showcase the practical benefits of the free-response paradigm, integrating theoretical insights with practical improvements in SNNs.

2.2 Fidelity-Entropy Loss Improves Learning in Moment Neural Network. We conduct numerical experiments using three well-established data sets: MNIST (Lecun et al., 1998), Fashion-MNIST (F-MNIST) (Xiao et al., 2017) and CIFAR10 (Krizhevsky, 2009). For MNIST and F-MNIST, our MNN implementation uses only one hidden layer, while for CIFAR-10, we employ a more complex model with three hidden layers. The architectural configurations are not intended to showcase training efficiency, but to thoroughly evaluate the models' ability to express prediction confidence across various task complexities, ranging from high-accuracy scenarios like MNIST to more challenging situations such as CIFAR-10 where performance is significantly constrained. We refer to models trained with and without the fidelity-entropy loss as WithFidelity and MeanOnly, respectively. In WithFidelity, we assign the weight coefficients as $[0.8, 0.2/(n-2), \ldots 0.2/(n-2)]$, where n represents the number of categories to be classified. This weight configuration emphasizes the need to optimize the prediction confidence between the highest and the second-highest entries in terms of the readout mean, and we find that it works quite well. Detailed descriptions of the experiments are in section 4.

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In our initial analysis, we focus on comparing the accuracy of the test set throughout the training period. In MNN models, since the readout mean can be obtained analytically, we can use it to determine the classification accuracy over the data set without considering the covariance. As illustrated in Figure 3a, models trained with fidelity-entropy loss (WithFidelity) and those trained without it (MeanOnly) show similar levels of accuracy throughout training epochs (refer also to Table 1). This result aligns with our theory, as the ability of making correct decisions is optimized by the cross-entropy term in equation 2.7, which is included in the loss function for training both WithFidelity and MeanOnly models. Significantly, it also indicates that additional constraints from the fidelity-entropy loss entail no performance compromises.

Instead of accuracy, the fidelity-entropy loss aims to optimize the model's confidence expression ability, though it does not ensure an optimized result across all input samples. There are three possible outcomes (see Figure 3b). In the most favorable scenario (left panel), WithFidelity efficiently shifts the correct prediction (red star) away from the decision boundary (dashed line), reducing covariance and aligning the major axis toward the boundary (solid ellipse), as opposed to the (MeanOnly) model. Conversely, misclassified samples (shown by a black cross) are moved toward the boundary with higher covariance (dashed ellipse). In the intermediate scenario (middle panel), the WithFidelity model may focus on optimizing either the mean or the covariance individually, but not both simultaneously, making it difficult to consistently enhance confidence expression. In the worst-case scenario (right panel), the fidelity-entropy loss can lead to misguided optimizations, thereby reducing confidence in accurate predictions while unjustifiably increasing confidence in incorrect ones.

To evaluate the impact of fidelity-entropy loss on model performance, we use area under the receiver operating characteristic (AUROC—AUC for short) analysis to evaluate the models' ability to distinguish between correct and incorrect predictions (Fawcett, 2006). Our study concentrated on five primary metrics: DV Mean, the difference between the two top entries in terms of the readout mean, which forms the numerator in equation 2.6; DV Std, using the same entry index as DV Mean, which corresponds to the denominator in equation 2.6; Fidelity, a combination of DV Mean and DV Std, defined in equation 2.6; Entropy, derived from readout covariance, as detailed in equation 4.15; and Softmax, the highest score obtained by applying the softmax function to the readout mean.

As demonstrated in Figure 3c, metrics that incorporate the readout mean (DV Mean, Fidelity, Softmax) are the most effective in distinguishing decision outcomes, with higher AUC indicating better performance. The upper bounds of these mean-related metrics are constrained by model classification accuracy, as higher accuracy leads to higher AUC. Given that both the WithFidelity and MeanOnly models can enhance the readout mean through the cross-entropy component, they achieve comparable AUC scores for

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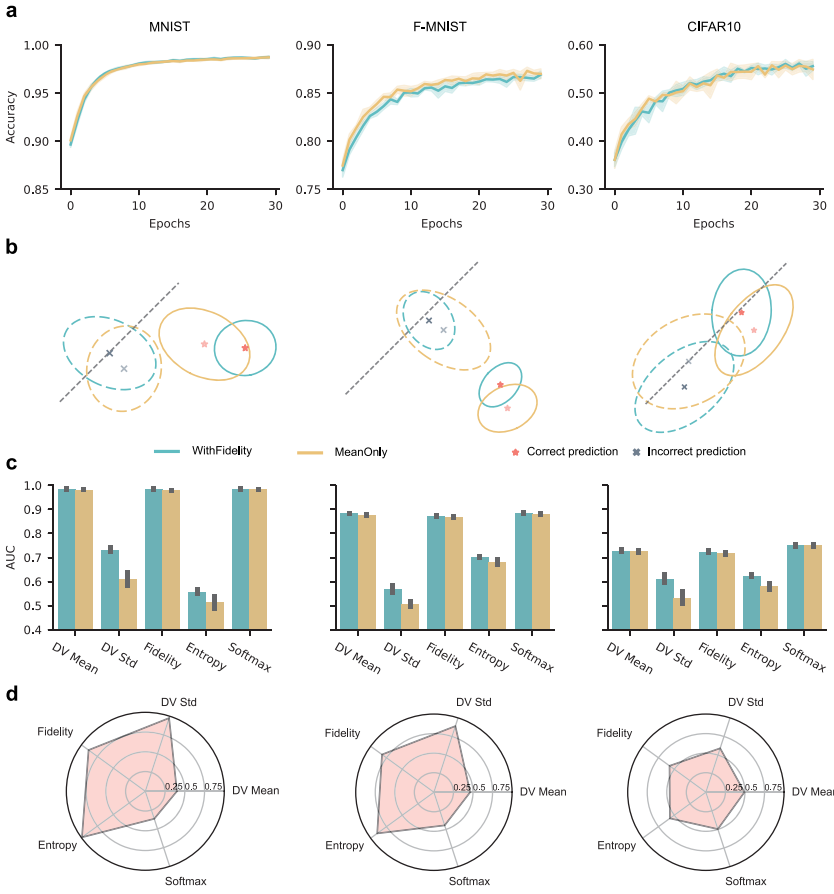


Figure 3: Efficacy of fidelity-entropy loss in training MNNs. (a) Accuracy of MNN models with and without fidelity-entropy loss across epochs, colored cyan and yellow, respectively. This color code is consistently used in the other plots. The shaded regions denote the 95% confidence intervals. (b) Case study examples showing the influence of the fidelity-entropy loss on model outputs. Correct predictions (red stars) and incorrect predictions (black crosses) are plotted based on the top two readout means, with their covariances represented by ellipses. Solid lines indicate covariances for correct cases, while dashed lines correspond to incorrect ones. The decision boundary is depicted as a gray dashed line. (c) AUC that evaluates the models' capability to differentiate correct from incorrect predictions using different confidence metrics. The error bars denote the 95% confidence intervals. (d) Proportion of samples in which a model can have an improved confidence expression using selected metrics after introducing the fidelity-entropy loss in training.

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these metrics, though the WithFidelity models tend to produce marginally superior results. The most significant advancements are seen in the DV Std and Entropy metrics, where WithFidelity models exhibit notably higher AUC compared to MeanOnly models. Namely, the fidelity-entropy loss mainly optimizes the output covariance structure of a model.

The following analysis examines the probability of observing an improved confidence expression under arbitrary input when the model is trained with the fidelity-entropy loss. For input samples that are accurately classified, metrics such as the DV mean, Fidelity, and Softmax are expected to rise, while the DV Std and Entropy are expected to fall. In contrast, for misclassified instances, the pattern is expected to reverse. A probability greater than 0.5 denotes a positive effect, while a value below 0.5 implies a negative effect. As depicted in Figure 3d, both DV Std and Entropy show marked enhancements across all data sets, underscoring the effectiveness of the fidelity-entropy loss in optimizing the readout covariance. In contrast, DV Mean and Softmax show a minor decrease, indicating a slight trade-off effect in introducing the proposed loss. Neverthless, because of the considerable improvement in DV Std, a notable increase in the metric Fidelity is observed, eliminating the adverse effect that comes from DV Mean. The extent of these improvements is mainly influenced by the model's accuracy, showing significant benefits in easy tasks like those trained on MNIST and F-MNIST, as opposed to CIFAR-10. This discrepancy arises because the optimization direction in the fidelity-entropy loss is contingent on the prediction outcome. In CIFAR-10, the nearly equal count of correct and incorrect predictions at each training step results in the mutual cancelation of optimizations from diverse inputs, leading to a minimal optimization effect.

2.3 Fidelity-Entropy Loss Leads to More Consistent Prediction and Shorter Convergence Time in SNNs. Leveraging the intrinsic link between MNN and SNN, we seamlessly transition to SNNs by reconstructing them using trained parameters of the MNN models discussed previously. Specifically, we use the MNN models depicted in Figure 3b as a basis for reconstruction, aiming to elucidate the enhancements that fidelity-entropy loss brings to SNNs. To rigorously evaluate performance, we conduct 100 trials for each sample in the test set and 100 milliseconds of simulation in each trial (corresponding to 100 time steps).

We first investigate how classification accuracy evolves as the readout time increases, represented by the solid lines in Figure 4a, with the cyan and yellow indicating the WithFidelity and the MeanOnly model, respectively. This color code is maintained throughout this work. While the accuracy differences between WithFidelity and MeanOnly SNN models are minimal at the end of the simulations for all three tasks (see Table 7), the trend of accuracy highlights the significance of the fidelity-entropy loss. In the simple task (MNIST), the fidelity-entropy loss shows no clear benefit, with both

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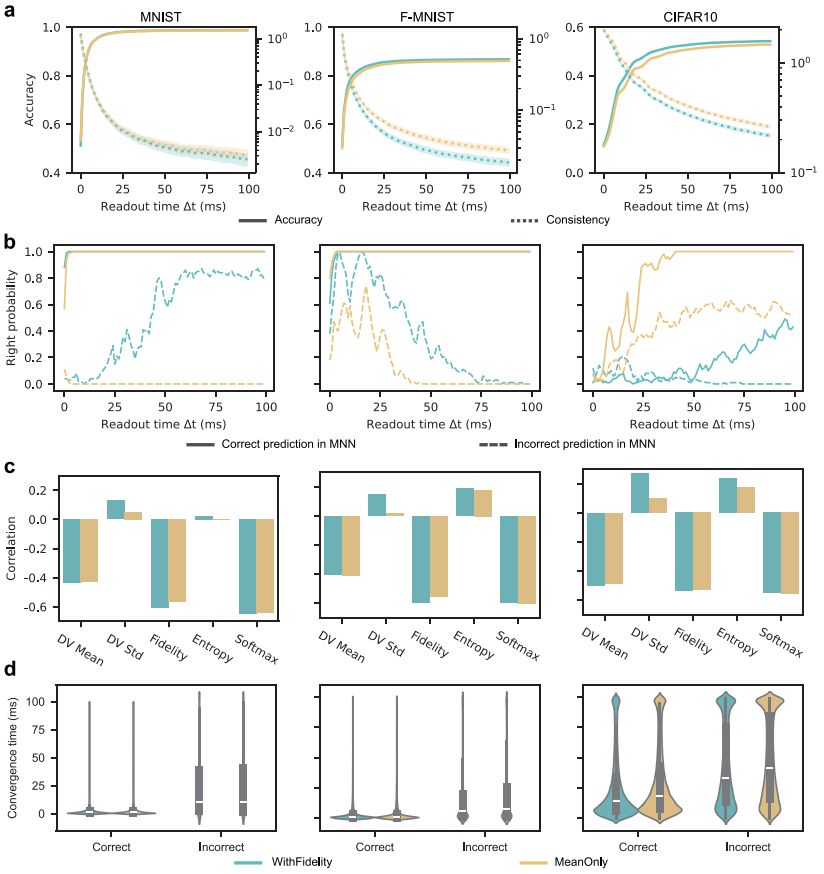


Figure 4: Training with the fidelity-entropy loss enhances prediction consistency and reduces inference latency. (a) Model performance as a function of readout time, contrasting accuracy (solid line) and consistency entropy (dotted line) between models trained with and without the fidelity-entropy loss. The shaded areas denote the 95% confidence intervals. (b) Case study of how the probability of correct predictions evolves during the readout period in different models, employing identical input samples as in Figure 3b. Solid and dashed lines represent correct and incorrect classifications by the respective MNN models. (c) Pearson correlation coefficients between the convergence time of SNNs and the confidence metrics of the corresponding MNNs. The convergence time of SNNs is defined as the time after which the SNN's prediction remains stable. (d) Distributions of SNN convergence time categorized by correct and incorrect predictions. In all plots, we use cyan and yellow to distinguish WithFidelity and MeanOnly models, respectively.

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models quickly attaining high accuracy. However, in more complex tasks, the WithFidelity model achieves a faster improvement in accuracy in the early stages compared to the MeanOnly model. Although this effect is minor in F-MNIST, it becomes more noticeable in CIFAR-10, suggesting the efficacy of fidelity-entropy loss in more complex tasks.

This phenomenon can be attributed to differences in prediction consistency, which is evaluated by the entropy of the probability distribution of the categories selected by the model during simulation (see equation 4.16) in section 4). A lower entropy indicates more consistent predictions among different trials and vice versa. Similar to the accuracy trend, the entropy for prediction distribution in both models diminishes with increasing readout time (the dotted lines in Figure 4a). In particular, WithFidelity SNN exhibits a faster decrease in entropy compared to MeanOnly SNN. This distinction becomes particularly pronounced in F-MNIST and CIFAR-10 but is almost negligible in MNIST.

The observed improvements come from the optimized readout structures after introducing the fidelity-entropy loss. To demonstrate it, we evaluate how the probability of correct prediction evolves during simulations in different models, using the same samples illustrated in Figure 3b as inputs (see Figure 4b). In the optimal case with well-optimized mean and covariance (left panel), the sample that is correctly classified by the WithFidelity MNN model shows a higher initial probability of correct prediction and a slightly faster convergence in the corresponding WithFidelity SNN model (solid lines), as compared to their MeanOnly counterparts. For the sample that both MNN models misclassify, the probability of making a correct prediction in the MeanOnly SNN exhibits a rapid decline (yellow dashed line), indicating an undue confidence in the erroneous prediction. In contrast, since the readout distribution in the WithFidelity MNN occupies a larger portion of the left side of the decision boundary than that in the MeanOnly MNN, the WithFidelity SNN is more likely to pick the opposite (correct) prediction, represented as the probability of making a correct decision increases unexpectably in this input example (depicted by the cyan dashed line). When only the mean or covariance is optimized (middle panel), the WithFidelity SNN cannot ensure improvements in prediction dynamics compared to the MeanOnly SNN. In this case, the WithFidelity SNN may have a lower probability of making the correct prediction initially (solid lines) and alternatively can have a prolonged period that struggles to avoid consistently making the wrong prediction (dashed lines). In the worst-case scenario (right panel), when the corresponding MNN models achieve correct classifications, the WithFidelity SNN seldom attains a probability of correction prediction above 50%, in contrast to the MeanOnly SNN, which achieves 100% accuracy. Furthermore, in cases where both MNN models are misclassified, the WithFidelity SNN shows a rapid convergence to an incorrect result, while the MeanOnly SNN still has a reasonable chance of producing a correct prediction.

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These case studies support our theory that the decision-making dynamics in SNNs is primarily influenced by the geometric interaction between the readout distribution and the decision boundary, the former of which can be captured by the MNN's output. We thus investigate this association by calculating the Pearson correlation between the convergence time of SNNs' predictions and the confidence metrics obtained from the corresponding MNN outputs. In this context, convergence time refers to a specific moment in time when the SNN's prediction stabilizes and does not alter thereafter. If the simulation fails to identify such a moment, the final time step is considered the convergence time. As illustrated in Figure 4c, it is evident that the convergence time demonstrates a negative correlation with metrics such as DV Mean, Fidelity, and Softmax, and a positive correlation with DV Std and Entropy, in agreement with our theoretical prediction (see equation 2.9). Significantly, the correlation coefficients for mean-related metrics are higher than those for covariance-related ones, underscoring the mean's major influence on convergence time. Moreover, the correlations for DV Std and Entropy are stronger when the model is trained with fidelity-entropy loss, illustrating this loss function's effectiveness in improving the covariance structure of readouts. Furthermore, the correlation between covariance-related metrics (DV Std and Entropy) and convergence time is more substantial in CIFAR-10 than in MNIST and F-MNIST, suggesting that covariance has a more critical impact on convergence time in more difficult tasks where model performance is limited.

Considering that the confidence metrics obtained from MNNs are indicative of the correctness of a model prediction (see Figure 3c), the convergence time of SNNs should similarly discern correctly and incorrectly classified samples, given the correlation with the MNN metrics. Consequently, we examined the distributions of convergence time of correct and incorrect predictions, respectively. The violin plots (see Figure 4d) display that correctly predicted samples typically converge more quickly, with their convergence time distributions having a positive skew. In contrast, misclassified instances require more time to converge, presenting a wider distribution. Furthermore, WithFidelity SNN demonstrates a faster convergence time in comparison to the MeanOnly SNN, especially evident in the more challenging task CIFAR-10. Due to the differences in convergence times among samples, the traditional interrogation paradigm, which uses a constant readout duration for all samples, must set this duration based on the longest convergence time of a correctly classified sample for optimal performance. However, since many samples converge more quickly, this approach leads to energy inefficiency. Thus, adaptive stopping techniques are needed to enhance energy efficiency.

2.4 Efficiency and Confidence Expression in SNNs through Adaptive Stopping. Contrary to abstract models like DDMs, which come with predefined evidence distributions facilitating the setting of an optimal decision

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threshold, real-world tasks pose challenges in identifying the distribution of instantaneous evidence. Although precise estimation of these distributions can be achieved through extended sampling periods or ensemble methods (Legenstein & Maass, 2014), these strategies conflict with the main objective of the free-response paradigm, which aims to save time and energy adaptively. Therefore, it is essential to devise an efficient stopping policy that maximizes the utilization of available evidence collected with limited time.

In this study, we suggest that using the difference between the largest and second-largest readout units can effectively signal a reasonable stopping point. When the difference increases, the probability of altering the existing decision will decrease, since fluctuations in the instantaneous readout $\mathbf{r}$ will not affect the relative ranking of the accumulated evidence $\mathbf{d}$ collected so far. In practice, we begin by sorting the cumulative readout $\mathbf{d}$ at each time step t in descending order to obtain $\tilde{\mathbf{d}}(t) = (\tilde{d}_1^t, \tilde{d}_2^t, \ldots, \tilde{d}_n^t)$, where $\tilde{d}_1^t$ is the highest value in $\mathbf{d}(t)$, $\tilde{d}_2^t$ is the second highest, and so on. According to this policy, the inference process will stop once the quantity $\tilde{d}_1^t - \tilde{d}_2^t$ exceeds a predefined threshold Θ_{Diff}. Furthermore, once this quantity meets the stopping criterion, it is reasonable to conclude that all remaining readout units are unlikely to exceed the largest one in the future, thus simplifying the stopping problem in multichoice tasks.

We refer to this policy as the Difference policy and compare it to various alternative policies: Fixed, Max, Linear, and Oracle. For the Fixed policy, it stops the inference process at a predetermined time regardless of evidence gathered so far, which serves as a lower bound of efficiency that arbitrary stopping policies should have. The Max policy stops inference once the highest readout unit surpasses a certain threshold ($\tilde{d}_1^t > \Theta_{\text{Max}}$). The Linear policy involves training a logistic classifier to distinguish between converged and unconverged predictions based on sorted readouts, stopping the inference once the classifier's output score surpasses a predetermined threshold. Initially, we collect the sorted cumulative readouts $\tilde{\mathbf{d}}(t)$ at all time points throughout the simulation, labeling the readouts before convergence as 0 and the rest as 1. We then train a logistic classifier f to differentiate $\tilde{\mathbf{d}}(t)$. The Linear policy uses this classifier to stop the inference process when $f(\tilde{\mathbf{d}}(t)) > \Theta_{\text{Lin}}$. Consequently, the Linear policy helps estimate the maximum efficiency for stopping policies using linear methods. Note that both the Difference and Max policies can be viewed as specific implementations of the Linear policy. The Oracle policy serves as an optimal case in which inference halts exactly on reaching convergence. That is, the stopping time in this policy matches the convergence time precisely, providing a theoretical upper bound for the efficiency that any stopping policy can achieve.

We compare the performance of various stopping policies with different decision thresholds. In the following, we present results on the WithFidelity model (see Figure 5a). For this plot, similar trends are found for the MeanOnly model (see Figure 7). The Oracle policy (orange star) reaches peak accuracy in the shortest time, so proximity to this policy indicates

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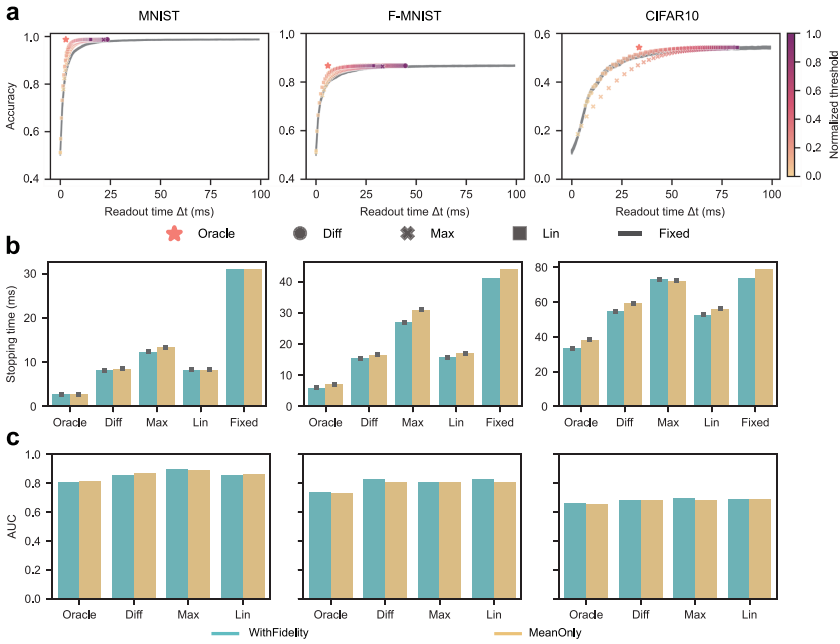


Figure 5: Efficiency of different stopping policies in SNNs. (a) Accuracy-speed trade-offs of stopping policies with different decision thresholds. The Oracle policy (orange star) serves as the efficiency upper bound that an ideal policy can have. Accuracy-readout time trade-offs are depicted for the Difference (Diff), Max, and Linear (Lin) policies with circles, crosses, and squares, respectively, each colored according to the normalized thresholds they used. The Fixed policy (gray line) is provided for baseline comparison. (b and c) Cyan denotes the WithFidelity model results and yellow represents the MeanOnly model. (b) Average readout time for models equipped with different stopping policies to achieve the same accuracy as the Oracle policy (0.005 tolerance). The error bars present the 95% confidence intervals. (c) AUC scores of using stopping time to distinguish correct and incorrect predictions. The stopping time is obtained by stopping policies with the same decision thresholds used in panel b. In panels b and c, cyan denotes the WithFidelity model results, while yellow represents the MeanOnly model.

superior performance. For simple tasks like MNIST and F-MNIST, the Difference, Max, and Linear policies outperform the Fixed policy, achieving higher accuracy in less time. The Linear policy slightly outperforms others in short readout durations. As decision thresholds increase, these policies approach Oracle-like accuracy, balancing accuracy and time efficiency. In the more complex CIFAR-10 task, the Max policy underperforms as compared to the Fixed policy under time constraints. Conversely, the

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Difference and Linear policies excel, achieving higher accuracy more efficiently, demonstrating their flexibility and capability for rapid convergence. While the Linear policy marginally outperforms the Difference policy when speed is emphasized, it requires prior training of a suitable classifier, making the Difference policy advantageous due to its simplicity in setting a decision threshold.

To quantify the effectiveness of various stopping strategies, we compare the minimum average readout time used for each policy to achieve Oracle-level accuracy (0.005 tolerance). Figure 5b demonstrates that WithFidelity and MeanOnly models benefit from an enhanced time efficiency due to the implementation of stopping policies, suggesting that these policies operate independent of the model training approach. The Linear policy only demonstrates slightly higher efficiency than the Difference policy, indicating that the efficiency of the Difference policy approaches the upper limit of linear stopping policies. Conversely, while the Max policy shortens the required readout time when compared to the Fixed policy, it still requires more time than both the Difference and Linear policies. Specifically, the Max policy provides minimal time-saving advantages in CIFAR-10, suggesting its inefficiency in difficult tasks where model performance is restricted. This highlights the reliability of our Difference policy, which remains as effective across various tasks as the Linear policy, even though there is a gap from the Oracle policy. We also verify the effectiveness of our Difference policy in SNN models trained using ANN-to-SNN conversion techniques (Hao et al., 2023; Li et al., 2022), as detailed in Table 3.

Similar to the experiment shown in Figure 3c, we also evaluate how well the stopping time can be used to indicate the correctness of a prediction. Figure 5c shows that the AUC for all policies in the three tasks is above 0.5, regardless of training methods, demonstrating that stopping time can serve as an indicator for prediction confidence, where quicker stops are more likely to be correct predictions and vice versa. However, the AUCs derived from this metric are largely influenced by the complexity of the task (reflected in the model's classification accuracy), with only slight variations between different policies and models. Notably, the Oracle policy that stops inference optimally shows a lower AUROC compared to other policies. This is because the limited temporal resolution causes the convergence time of both correct and incorrect predictions to coincide, making them hard to differentiate. In contrast, an imperfect stopping policy requires a higher stopping threshold to achieve desired accuracy, similar to a high decision threshold in equation 2.9, which results in more distinct stopping time distributions for correct and incorrect predictions.

3 Discussion

In this work, we have formulated a comprehensive theory that delineates the decision-making mechanisms within SNNs. This theory breaks down

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decision making into two key components: identifying the optimal choice based on current evidence and assessing its optimality if further evidence is obtained. We use them as guiding principles to refine SNN implementations across training and inference phases. During the training phase, we have presented a new training objective that motivates the SNN not only to learn to make the correct decisions but also to shape its decision confidence with respect to the outcomes, leading to reduced trial-to-trial variability and faster inference speed. During the inference phase, we have introduced stopping policies to mimic the dynamic stopping mechanism used in DDMs. Through numerical experiments, we have demonstrated that the difference between the largest and the second largest entries in cumulative readouts is highly informative in determining whether an inference should be stopped. SNNs equipped with such adaptive stopping mechanisms can transition from the commonly used interrogation paradigm to the more flexible free-response paradigm, leading to a further improvement in energy efficiency. Furthermore, our work reveals an inverse correlation between stopping time and the prediction's correctness. Specifically, predictions with short stopping times tend to be correct, corroborating findings from previous psychophysical studies (Gold & Shadlen, 2007; Kepecs et al., 2008; Kiani et al., 2014).

For both biological and artificial neural systems, maintaining a well-calibrated confidence expression ability is essential for steering future actions, as it aids in recognizing uncertainty associated with decision making. The predominant neuroscience theories posit that confidence is equal to the probability that a decision is correct (Kepecs & Mainen, 2012; Mamassian, 2016; Pouget et al., 2016). However, our experimental findings suggest that the ability to express confidence is greatly influenced by the accuracy of the model. This means that uncertainty in decision making should be categorized into two types: objective uncertainty and subjective uncertainty. Objective uncertainty corresponds to the frequency that the decision made by a neural system is correct, while subjective uncertainty corresponds to the degree to which the system believes its decision to be correct. These two forms of uncertainty may align if the neural system has fully grasped the true posterior probability of the selected option, a feat that is almost unattainable in practical scenarios (Beck et al., 2012). The absence of a definitive posterior probability for decisions also poses a significant challenge in improving the confidence calibration of models in artificial intelligence (Gawlikowski et al., 2023). Therefore, since we care more about whether the system's outcome is trustworthy, it is crucial to take both the neural system's performance and its ability to express confidence into consideration before accepting its decision.

The examination of confidence in binary decision-making tasks is prevalent in neuroscience, but real-world practices generally involve several potential options. Our theory introduces a nuanced perspective, positing that confidence arises from the disparity between the first- and

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second-best choices. This difference becomes especially pronounced in multichoice contexts, where the traditional view and ours diverge, particularly under circumstances of low confidence. The conventional perspective interprets confidence as the posterior probability of choice, whereas our theory underscores the direct comparison between the top two options. In instances where the top two choices are almost equally probable and differ considerably from the rest, our framework and traditional models may yield divergent confidence expression. This finding is consistent with the results of Li & Ma (2020) in a three-option visual categorization task, highlighting the nuanced nature of confidence expression.

In practice, it is unnecessary to wait until a system's decision becomes unchanged, since our primary focus is on the selection itself rather than the exact posterior probability associated with that option. Consequently, a mechanism is required that halts the inference process. Adaptive stopping is a concept with a rich historical background that can trace its roots to sequential analysis, a mathematical framework for adaptive decision rules developed during World War II (Wald, 1973). This framework has become a crucial element of statistical decision theory (Poor, 1994), highlighting the ability of adaptive decision making to improve any readout scheme through continuous real-time monitoring and adjustments (D'Anjou et al., 2016). Despite its established significance, the application of adaptive decision making within the realm of neuromorphic computing has not been explored much (Li, Jones et al., 2023; Li et al., 2024). In this regard, our research aims to connect this divide by utilizing the extensive knowledge of sequential analysis and adaptive decision theories to explore the spatio-temporal patterns of SNNs.

Although our findings offer valuable insights, a number of limitations merit attention. Our results are demonstrated on the moment neural network (MNN) framework (Feng et al., 2006; Lu et al., 2010; Qi et al., 2023), which, despite its utility, faces scalability challenges that hinder the training of more sophisticated networks for complex tasks. Furthermore, the fidelity-entropy loss that we have introduced may not seamlessly integrate into the training of SNNs via conventional ANN-SNN conversion methods due to the inability of ANNs to mimic the covariance propagation characteristic of MNNs. Future efforts could explore incorporating fidelity-entropy loss in direct SNN training methods such as surrogate gradient, allowing for the approximation of mean and covariance through sampling techniques. Furthermore, in this work, the decision thresholds used in the stopping policies during the inference stage are fixed to enable a comparative analysis of the efficiency of various policies. However, a threshold that adapts based on the accuracy of recent decisions may better represent the fluctuating decision-making mechanisms seen in biological organisms (Deco et al., 2009; Gupta et al., 2024; Vickers & Packer, 1982). This self-adjusting mechanism has the potential to allow SNNs to automatically determine the best thresholds, overcoming the difficulties of manual calibration in

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unpredictable input situations and when deploying SNNs on various hardware platforms. In the future, an in-depth integration of cognitive neuroscience and brain-inspired computing can facilitate a deeper understanding of neural computation and offer a more thorough investigation of the potential of SNNs.

4 Methods

4.1 Leaky Integrate-and-Fire Neuron Model. We used the leaky integrate-and-fire (LIF) spiking neuron model whose membrane potential dynamics is described by

$$\frac{dV_i}{dt} = -LV_i(t) + I_i(t), \tag{4.1}$$

where the subthreshold membrane potential $V_i(t)$ of a neuron i is driven by the total synaptic current $I_i(t)$ and $L = 0.05\ \text{ms}^{-1}$ is the leak conductance. When the membrane potential $V_i(t)$ exceeds a threshold $V_{\text{th}} = 20\,\text{mV}$, a spike is emitted, as represented by a Dirac delta function. Afterward, the membrane potential $V_i(t)$ is reset to the resting potential $V_{\text{res}} = 0$ mV, followed by a refractory period $T_{\text{ref}} = 5$ ms. The synaptic current takes the form

$$I_i(t) = \sum_j w_{ij} S_j(t) + I_i^{\text{ext}}(t), \tag{4.2}$$

where $S_j(t) = \sum_k \delta(t - t_j^k)$ represents the spike train generated by presynaptic neurons.

A final output $\mathbf{y}$ is read out from the spike count $\mathbf{n}(\Delta t)$ of a population of spiking neurons over a time window of Δt as

$$y_i(\Delta t) = \frac{1}{\Delta t} \sum_j w_{ij}^{\text{out}} n_j(\Delta t) + \beta_i, \tag{4.3}$$

where w_{ij}^{out} and β_i are the weights and biases of the readout, respectively. A key characteristic of the readout is that its variance decreases as the time window Δt increases.

4.2 Moment Neural Network. The moment embedding approach (Feng et al., 2006; Lu et al., 2010; Qi et al., 2023) begins with mapping the fluctuating activity of spiking neurons to their respective first- and second-order moments,

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$$\mu_i = \lim_{\Delta t \to \infty} \frac{\mathbb{E}[n_i(\Delta t)]}{\Delta t} \tag{4.4}$$

and

$$\Sigma_{ij} = \lim_{\Delta t \to \infty} \frac{\mathrm{Cov}[n_i(\Delta t), n_j(\Delta t)]}{\Delta t}, \tag{4.5}$$

where $n_i(\Delta t)$ is the spike count of neuron i over a time window Δt. In practice, the limit of $\Delta t \to \infty$ is interpreted as a sufficiently large time window relative to the timescale of neural fluctuations. We refer to the moments μ_i and Σ_{ij} as the mean firing rate and the firing covariability in the context of MNN, respectively.

For the LIF neuron model, equation 4.1, the statistical moments of the synaptic current are equal to (Feng et al., 2006; Lu et al., 2010)

$$\begin{cases} \bar{\mu}_i = \sum_k w_{ik}\mu_k + \hat{\mu}_i^{\mathrm{ext}}, & (4.6) \\ \bar{\Sigma}_{ij} = \sum_{kl} w_{ik}C_{kl}w_{jl} + \hat{C}_{ij}^{\mathrm{ext}}, & (4.7) \end{cases}$$

where w_{ik} is the synaptic weight and $\hat{\mu}_i^{\mathrm{ext}}$ and $\hat{\Sigma}_{ij}^{\mathrm{ext}}$ are the mean and covariance of an external current, respectively. Note that from equation 4.7, it becomes evident that the synaptic current is correlated even if the presynaptic spike trains are not. Next, the first- and second-order moments of the synaptic current are mapped to those of the spiking activity of the postsynaptic neurons. For the LIF neuron model, this mapping can be obtained in closed form through a mathematical technique known as the diffusion approximation (Feng et al., 2006; Lu et al., 2010) as

$$\begin{cases} \hat{\mu}_i = \phi_\mu(\bar{\mu}_i, \bar{\sigma}_i), & (4.8) \\ \hat{\sigma}_i = \phi_\sigma(\bar{\mu}_i, \bar{\sigma}_i), & (4.9) \\ \hat{\rho}_{ij} = \chi(\bar{\mu}_i, \bar{\sigma}_i)\chi(\bar{\mu}_j, \bar{\sigma}_j)\bar{\rho}_{ij}, & (4.10) \end{cases}$$

where the correlation coefficient $\hat{\rho}_{ij}$ is related to the covariance as $\hat{\Sigma}_{ij} = \hat{\sigma}_i\hat{\sigma}_j\rho_{ij}$. The mapping given by equations 4.8 to 4.10 is called *moment activation*, which is differentiable so that gradient-based learning algorithms can be implemented and the learning framework is known as the moment neural network (MNN) (Qi et al., 2023).

The functions ϕ_μ and ϕ_σ are derived from the LIF neuron model through a Fokker-Planck formalism, and they together map the mean and variance of the input current to those of the output spikes according to Capocelli and Ricciardi (1971), Feng et al. (2006), and Lu et al. (2010),

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$$\begin{cases} \mu = \dfrac{1}{T_{\text{ref}} + \frac{2}{L}\int_{I_{\text{lb}}}^{I_{\text{ub}}} g(x)dx}, & (4.11) \\ \sigma^2 = \dfrac{8}{L^2}\mu^3 \displaystyle\int_{I_{\text{lb}}}^{I_{\text{ub}}} h(x)dx, & (4.12) \end{cases}$$

where T_{ref} is the refractory period with integration bounds $I_{\text{ub}}(\bar{\mu}, \bar{\sigma}) = \frac{V_{\text{th}}L - \bar{\mu}}{\sqrt{L}\bar{\sigma}}$ and $I_{\text{lb}}(\bar{\mu}, \bar{\sigma}) = \frac{V_{\text{res}}L - \bar{\mu}}{\sqrt{L}\bar{\sigma}}$. The constant parameters L, V_{res}, and V_{th} are identical to those in the LIF neuron model in equation 4.1. The pair of Dawson-like functions $g(x)$ and $h(x)$ appearing in equations 4.11 and 4.12 are $g(x) = e^{x^2}\int_{-\infty}^{x} e^{-u^2}du$ and $h(x) = e^{x^2}\int_{-\infty}^{x} e^{-u^2}[g(u)]^2du$. The function χ, which we refer to as the linear response coefficient, is equal to $\chi(\bar{\mu}, \bar{\sigma}) = \frac{\bar{\sigma}}{\sigma}\frac{\partial\mu}{\partial\bar{\mu}}$ and is derived using a linear perturbation analysis around $\bar{\rho}_{ij} = 0$ (Lu et al., 2010). This approximation is justified because pairwise correlations between neurons in the brain are typically weak. Numerical simulations also show that the linear approximation works reasonably well for most inputs, even when the input correlation is away from zero (Qi, 2024). An efficient numerical scheme is used to evaluate moment activation and its gradients (Qi, 2024).

Derived from the spiking neuron model on a mathematically rigorous ground, the moment neural network faithfully captures spike count variability up to second-order statistical moments (Feng et al., 2006; Lu et al., 2010) and can be considered as a minimalistic yet rich description of the statistical properties of pairwise neural interactions. Meanwhile, MNN retains the analytical tractability and differentiability of continuous rate models with which gradient-based learning can be performed. The network parameters trained in this way can then be used directly to recover the spiking neural network without fine-tuning of free parameters.

4.3 Model Setup for MNN Training. MNNs were trained on three data sets: MNIST (Lecun et al., 1998), Fashion-MNIST (F-MNIST) (Xiao et al., 2017), and CIFAR-10 (Krizhevsky, 2009). For MNIST and F-MNIST, the MNN architecture included one hidden layer, while for CIFAR-10, it comprises three hidden layers, each containing 1000 LIF neurons in a feedforward configuration.

The MNIST data set features 60,000 training and 10,000 test images of handwritten digits, whereas Fashion-MNIST provides 60,000 training and 10,000 testing images of grayscale clothing items. CIFAR-10 offers a collection of natural images, with 50,000 for training and 10,000 for validation. The input dimension for the MNNs corresponds to the number of pixels in each image (784 for MNIST and F-MNIST, 3072 for CIFAR-10).

Input images $\mathbf{x}$ were encoded using a Poisson-rate scheme, with mean μ and variance σ^2 proportional to pixel intensities with a gain factor of 1. The correlation coefficients of the input were set to zero. Given that all data sets

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encompass 10 categories, a linear readout translated the moments of the last hidden layer into moments in a 10-dimensional space for classification using the operations in equations 4.6 and 4.7, which were the outputs of the MNN.

Using the MNN output (mean $\hat{\mu}$ and covariance $\hat{\Sigma}$), we applied the loss function described by equation 2.8 and compared models trained with (WithFidelity) and without (MeanOnly) the fidelity-entropy loss. Crucially, the fidelity-entropy loss calculation involved sorting $\hat{\mu}$ and $\hat{\Sigma}$ in descending order based on $\hat{\mu}$. Let $\tilde{\mu} = (\tilde{\mu}_1, \tilde{\mu}_2, \ldots, \tilde{\mu}_n)$ represent the sorted readout mean, where $\tilde{\mu}_1$ is the highest value in $\hat{\mu}$, $\tilde{\mu}_2$ is the second-highest value, and so on. By building the index mapping between $\tilde{\mu}$ and $\hat{\mu}$, each pair of entries used to compute the fidelity-entropy loss can be related to the respective mean and covariance, based on the original indices of $\tilde{\mu}_1$ and the others.

For the default setting in the main text, the primary weight w_1 was set to 0.8, emphasizing the fidelity entropy between the top two entries, while the remaining pairs received a weight of $w_k = 0.2/(n-2)$, for all $k > 1$, where $n = 10$ is the number of categories to classify. In appendix B, we provided a further comparison of two alternative weight configurations. For the Uniform configuration, we assigned the weight $w_k = 1/(n-1)$ for each k. For the Trainable configuration, the weights w_k were optimized through learning, with $k = 1, 2, \ldots, 9$. Note that we cannot simply make w_k in the fidelity entropy loss in equation 2.8 a trainable parameter because we want to maximize rather than minimize w_k corresponding to the component contributing to the most to fidelity-entropy so that the model can be correctly optimized. Instead, we propose the following modification to equation 2.8 by adding a third loss term as

$$L(\theta, w_k) = L(\theta) + L(w_k). \tag{4.13}$$

The new loss term is equal to

$$L(w_k) = f(\text{label}, i)(-w_j), \tag{4.14}$$

where w_j is the weight in w_k that the corresponding entropy component $[-P_{ik} \log P_{ik} - (1 - P_{ik}) \log(1 - P_{ik})]$ is the largest. In particular, we applied the softmax function to w_k when using them in $L(\theta)$ to ensure that the weights were nonnegative and were summed up to one. Moreover, the gradient of $L(\theta)$ with respect to w_k is discarded, resulting in w_k being optimized exclusively by $L(w_k)$. Note that the optimization direction of $L(w_k)$ was also affected by the indicator function $f(\text{label}, i)$ in equation equation 2.8.

For all experiments, the readout time Δt in the loss function was fixed at 1. We conducted training under the conditions WithFidelity and MeanOnly over five trials for each data set. Additional comparisons under the

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Uniform and Trainable conditions were carried out with only trial each. Each experiment consisted of 30 epochs and employed a batch size of 50. Random cropping was used for data augmentation during training in which images were first padded with two pixels on all sides, then randomly cropped back to their original size. The model parameters were optimized using the AdamW optimizer with a learning rate of 0.001 and a weight decay of 0.01.

4.4 Data Analysis in MNN Models. Upon obtaining the model's outputs $\hat{\mu}$ and $\hat{\Sigma}$, we determined its predictions by identifying the largest entry of $\hat{\mu}$. Metrics including DV Mean, DV Std, and Fidelity were directly derived from equation 2.6, using the top two components of $\hat{\mu}$ and $\hat{\Sigma}$ sorted according to $\hat{\mu}$. Entropy was calculated from $\hat{\Sigma}$ using the formula

$$H = \frac{1}{2}\ln((2\pi e)^n|\Sigma|), \tag{4.15}$$

where n is the dimension of Σ. The Softmax score was determined by applying the softmax function to $\hat{\mu}$ and selecting the largest entry (see equation 2.3). Subsequently, we categorized the test set samples on the basis of the models' predictions. We calculated their AUCs using Scipy's built-in functions to quantify the efficacy of these metrics in indicating decision correctness.

To evaluate the potential improvement to classifying any input using fidelity-entropy loss, we analyzed the proportion of test set instances in which performance metrics such as DV Mean, Fidelity, and Softmax were superior in models with fidelity as opposed to those without it, specifically for correctly classified instances. Conversely, these metrics were expected to be lower for misclassified instances. The same comparison was conducted for DV Std and Entropy that these two metrics are expected to be smaller for correctly classified instances and vice versa.

4.5 Experimental Setup for SNN Simulation. The trained MNN parameters were used to reconstruct the SNN models simply by replacing the moment activation layer in MNN with LIF neurons, as explained in our previous study (Qi et al., 2023). The MNN models trained under the With-Fidelity and MeanOnly conditions, as shown in Figure 3b, were used to reconstruct the respective SNNs. These SNNs were referred to as the With-Fidelity SNN and MeanOnly SNN, respectively, and were also used in the comprasions in appendix B.

The input spike trains for these SNNs were created using a Poisson process with rates proportional to the pixel intensities of the images in the test data sets with a gain factor of 1. The evaluation of each instance was carried out within a time frame of 100 ms and a time resolution of $\delta t = 1$ ms, resulting in 100 instantaneous readout values for each instance.

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The starting membrane potentials of LIF neurons were randomly assigned from a uniform distribution ranging from V_{res} to V_{th}. Throughout the simulation, the membrane potentials were not reinitialized after processing each sample in a test set. This methodology was consistently applied to all tasks, and the simulation was replicated 100 times with different initial membrane potentials. The resulting instantaneous readout values $\mathbf{r}$ from these simulations were then examined for additional analysis.

4.6 Data Analysis in SNN Simulations. To analyze the readouts from the SNN simulations, we first aggregated the instantaneous readout values $\mathbf{r}(t)$ over time to calculate the cumulative readout $\mathbf{d}(t)$. For each sample image, the prediction of the model y_{pred} was obtained by identifying the index of the largest entry in $\mathbf{d}(t)$, and by comparing it with the ground truth at each time point t over repeated trials, we obtained the probability of correct predictions. Prediction consistency at time t was quantified using the entropy

$$H(t) = -\sum_k P^{(t)}(y_{\text{pred}} = k) \log P^{(t)}(y_{\text{pred}} = k), \tag{4.16}$$

where $P^{(t)}(y_{\text{pred}} = k)$ denotes the probability of the model predicting category k at time t, estimated by counting the occurrences of each predicted category in 100 trials. We computed the mean and confidence intervals of both the probability of correct predictions (i.e., accuracy) and the prediction consistency over all samples in the test set.

For the case studies depicted in Figure 4b, we reused the samples in Figure 3b as inputs, assessing the probability of making a correct prediction in WithFidelity and MeanOnly SNNs.

The convergence time from these simulations was defined as the earliest moment in time t^* when the prediction of a model remained constant within the simulation period ($t = 100$ ms). This procedure yielded a convergence time for each sample and trial. Subsequently, we examined the Pearson correlation between these convergence times and the five metrics previously derived from the corresponding MNN models for the input samples (see Figure 4c). Furthermore, by categorizing the convergence times based on the results of the corresponding predictions (correct or incorrect), we visualized the distribution of convergence times (see Figure 4d).

4.7 The Implementation of Stopping Policies. We have described how different stopping policies halt the inference process in section 2.4. Specifically, for Linear policy, we trained a logistic classifier f with the built-in function in Scipy with default settings to differentiate $\tilde{\mathbf{d}}(t)$.

We systematically varied decision thresholds for each policy to cover the full spectrum from low to high accuracy. For the Difference and Max

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policies, Θ ranged from 10 to 100, while for the Linear policy, it ranged from -1.5 to 10. To maintain consistent color coding, the thresholds were normalized to the unit interval in the visualization. By changing the decision threshold, we found the minimal decision thresholds for each policy, by which the disparity between the classification accuracy achieved by a policy and that by the Oracle policy remains below 0.005. Concurrently, we derived the associated stopping times for each individual sample on the basis of the stopping criterion of each policy. We then performed an AUROC analysis on these stopping times to assess their effectiveness in indicating whether the model's predictions were correct or incorrect.

In further experiments (see Table 3), we employed the ANN-SNN conversion methodology presented by Hao et al. (2023) and Li et al. (2022) to show that an adaptive stopping mechanism can enhance the efficiency of SNNs, independent of the training method. When working with the MNIST data set, we replicated the experimental configuration as previously outlined and trained ANNs for 30 epochs. For CIFAR-100 (Krizhevsky, 2009), we followed the training configurations set by Hao et al. (2023). In the case of these two data sets, the conversion from ANNs to SNNs was carried out using the approach proposed by Hao et al. (2023), known as the residual membrane potential (SRP) method. When dealing with the ImageNet data set (Deng et al., 2009), we employed the model pretrained by Li et al. (2022) and implemented their light calibration pipeline (LCP). Importantly, during these experiments, the inputs were kept constant, aligned with the pixel intensities of the images, thus eliminating variability caused by noisy inputs. For MNIST and CIFAR-100, simulations were conducted in 100 steps, while ImageNet required 256 steps. Following each simulation, we implemented the Difference policy on the output of these converted SNNs, evaluating the time required to reach a final accuracy of 95% to illustrate the applicability of our theory in different training methodologies and network architectures.

Appendix A: Additional Analyses of Learning and Inference in MNN and SNN

A.1 Accuracy Comparison of MNN Models Trained with and without Fidelity-Entropy Loss. In the main text, we have shown how accuracy changes during training in the MNN with and without fidelity-entropy loss, as shown in Figure 3a. Table 1 summarizes the final performance of each model. Although the average accuracy remains largely unchanged, the fidelity-entropy loss significantly reduces training variability across multiple trails. In all tasks, WithFidelity has a narrower confidence interval than MeanOnly. This reduction is most evident in CIFAR-10, where the interval is halved for models trained without fidelity-entropy loss. These findings underscore the significance of fidelity-entropy loss in enhancing the

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Table 1: The Accuracy of MNNs Trained with and without Fidelity-Entropy Loss.

	MNIST	F-MNIST	CIFAR-10
WithFidelity	98.75% ± 0.08%	86.92% ± 0.28%	55.62% ± 0.73%
MeanOnly	98.72% ± 0.11%	87.01% ± 0.39%	54.93% ± 1.61%

Note: The accuracy is assessed by models that were trained for 30 epochs with a 95% confidence interval.

Table 2: Comparision of the Reconstructed SNNs' Accuracy at the Final Step.

Data set	Loss function	ACC at final step	ACC of the corresponding MNN
MNIST	WithFidelity	98.76%	98.80%
	MeanOnly	98.64%	98.67%
F-MNIST	WithFidelity	86.72%	86.79%
	MeanOnly	86.04%	86.19%
CIFAR-10	WithFidelity	54.22%	56.79%
	MeanOnly	52.77%	55.7%

Note: The accuracy of the reconstructed SNNs is evaluated by averaging the results of 100 trials.

robustness of MNN model training, as it mitigates the impact of random weight initialization on classification performance.

A.2 Accuracy Comparison of Corresponding SNN Models. In the main text, we showed in Figure 4a how accuracy during inference in SNN changes with readout time Δt. Here, we further compare the final performance of SNN models to their MNN counterparts under different training conditions. Table 2 shows that under the WithFidelity condition, the performance difference is slightly less than under the MeanOnly condition. For simple tasks (MNIST), the performance gap is 0.04% for WithFidelity and 0.03% for MeanOnly. For more complex tasks (F-MNIST), the gap expands to 0.07% and 0.15%, respectively. The most challenging task (CIFAR-10) poses a gap of 2.57% and 2.93%, respectively. Thus, using fidelity-entropy loss during training helps reduce performance losses due to reconstruction and simulation.

A.3 Analysis of Hidden Neuron Activity in the Case Studies. In the main text, we provided case studies of the SNN performance for specific input images, as shown in Figure 4b. Here, we illustrate the neural activity of the final hidden layer for these input images, as depicted in Figure 6. While it may be challenging to determine the model's predicted category solely from the raster plots, these plots do reveal distinct collective firing patterns among the hidden neurons, indicating correlated rather than independent

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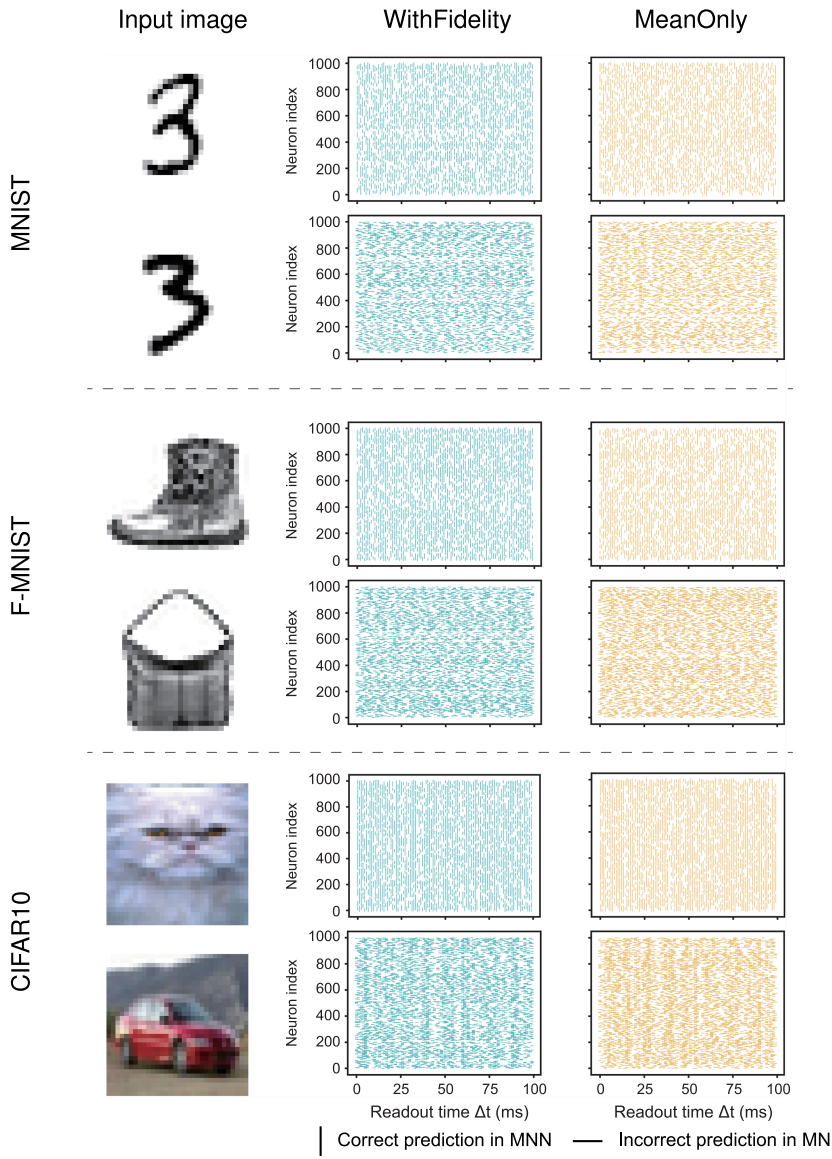


Figure 6: Raster plots of SNN spiking behavior in case studies. Input images are used in Figures 3b and 4b. The raster plots depict the spiking activity of the final hidden neurons in WithFidelity and MeanOnly SNNs. The rows illustrate different scenarios: the top row shows a sample correctly classified by both MNNs (vertical lines), while the bottom row shows a misclassified sample (horizontal lines).

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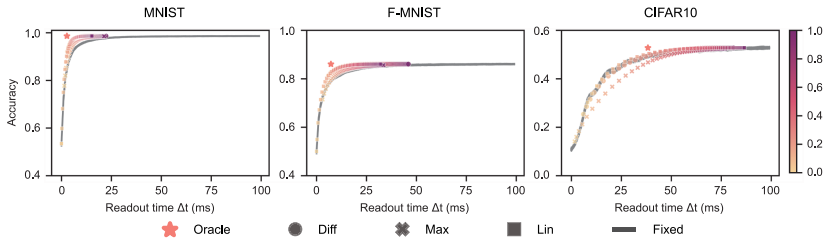


Figure 7: Speed-accuracy trade-off for different stopping policies applied on the MeanOnly SNN.

firing activities. Consequently, this correlated neural activity influences the decision-making dynamics during inference through the linear readout, as demonstrated in Figure 4b.

A.4 The Efficiency of Stopping Policies in MeanOnly SNN. Figure 5a in the main text demonstrates the performance of different stopping policies on the WithFidelity SNN. Here we conduct the same analysis on the MeanOnly SNN for comparison. As depicted in Figure 7, these policies exhibit similar behaviors in the MeanOnly SNN as well, suggesting that such adaptive stopping mechanisms can be applied to arbitrary SNNs. For both models, the performance of the proposed Difference policy has is comparable to that of the Linear policy. Although the Max policy can outperform the Fixed policy on MNIST and F-MNIST, it fails to show the efficiency of an adaptive stopping mechanism on the challenging CIFAR-10 task.

A.5 Application of the Stopping Policy to SNNs via ANN-to-SNN Conversion Techniques. To illustrate the general applicability of the stopping policies independent of the specific training method, we apply the Difference policy to SNN models obtained through ANN-to-SNN conversion techniques. Table 3 shows that the Difference policy achieves the same accuracy as the conventional Fixed policy more quickly, thereby enhancing energy efficiency.

Appendix B: Effect of Weight Configurations in Fidelity-Entropy Loss on Training

In the main text, we compared the training results with and without fidelity entropy loss (WithFidelity and MeanOnly). A potential variable affecting the training outcome is the weights w_k in the fidelity entropy assigned to the readout pairs. In the main text, we used default values of $w_1 = 0.8$ and $w_k = 0.2/(n-2)$ for all $j > 1$ (WithFidelity) to highlight the contribution of the pair consisting of the largest and second-largest entries. To further

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Table 3: Comparison of Average Steps Required with or without the Stopping Policy.

Data set	Method	Arch	ACC at final step	Average step (Oracle policy)	Average steps for 95% ACC	
					Fixed policy	Difference policy
MNIST	SRP (Hao et al., 2023)[a]	MLP	98.95%(T = 100)	4	4	2.89
CIFAR100	SRP (Hao et al., 2023)[a]	ResNet20	66.26%(T = 100)	9	11	10.65
ImageNet	LCP (Li et al., 2022)[b]	ResNet34	71.69%(T = 256)	117.45	189	153.26
ImageNet	LCP (Li et al., 2022)[b]	VGG16BN	50.75%(T = 256)	147.21	253	208.75

Notes: [a]Training from scratch.
[b]Using the pretrained models.

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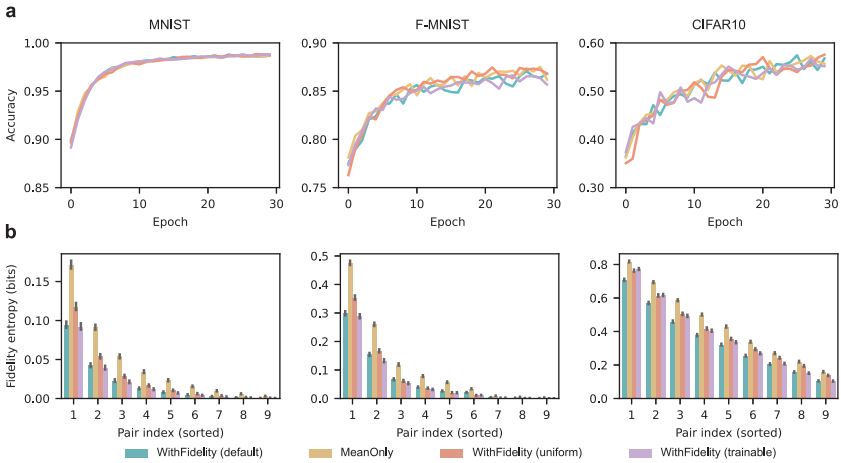


Figure 8: Impact of weight configurations on learning. (a) Accuracy on the test set over epochs. (b) Average fidelity-entropy loss contributed its components. The pair index is sorted so that pair 1 is about the top two entries, pair 2 is about the largest and third-largest entries, and so on. The error bars indicate the 95% confidence interval.

investigate the effect of different weight configurations, we consider two additional scenarios: one with $w_k = 1/(n-1)$ for all $j \geq 1$ (Uniform) and the other with trainable w_k (Trainable). For Trainable, w_k optimization tends to increase the weight corresponding to the highest fidelity-entropy loss contributor if the prediction turns out to be correct and decrease it otherwise (see section 4). We compare these scenarios (Uniform and Trainable) against MeanOnly and WithFidelity for both MNN and SNN models, maintaining consistency by reusing models from section 2.3 for WithFidelity and MeanOnly, while training Uniform and Trainable from scratch.

B.1 Effect of Fidelity-Entropy Weight Configurations on Training in MNN. In the main text, we used a default weight configuration to show that incorporating fidelity-entropy loss has minimal impact on a model's classification performance despite the additional constraints. Here, we provide additional analysis with different weight configurations. Figure 8a shows that models trained with different weight configurations for the fidelity-entropy loss have similar trends in the accuracy as the MeanOnly model. This observation corroborates our previous conclusion that the fidelity-entropy loss does not influence the classification accuracy a model can achieve. In the simple MNIST task, all models' accuracy steadily increases over training epochs, consistently showing comparable levels of performance. As the difficulty of tasks increases (F-MNIST and CIFAR-10),

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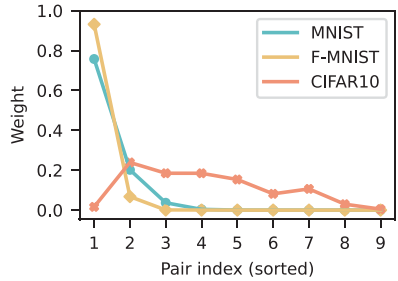


Figure 9: The obtained weight coefficients under the Trainable configuration.

their performance shows variability during training but stays within a comparable range.

We analyze the fidelity-entropy loss of each readout pair after training in various models (see Figure 8b). On average, the first pair (the largest and second-largest entries) contributes the most to the loss, with the gap between the first pair and the rest narrowing when task difficulty increases. In comparison to the MeanOnly model, incorporating fidelity-entropy loss effectively decreases the loss, irrespective of the weight configuration. Among the three configurations, the WithFidelity configuration used in the main text is the most robust to reducing the fidelity-entropy. While Trainable slightly outperforms-WithFidelity in MNIST and F-MNIST, it falters with CIFAR-10, where model performance is heavily constrained by task difficulty. Uniform is least effective compared to WithFidelity (default weight configuration) and Trainable across all tasks.

To understand why Trainable falls behind WithFidelity in CIFAR-10, we present the weight coefficients obtained from the Trainable model (see Figure 9). In the MNIST and F-MNIST data sets, the weight of the first pair dominates the fidelity-entropy loss, affirming our hypothesis that the disparity between the largest and second-largest entries of the readout is crucial for prediction confidence. In contrast, the weights trained on the CIFAR-10 task are more broadly distributed, with the weight of the first pair nearly equal to zero. Given that the ordering of the pair index for the fidelity-entropy loss is based on the readout mean, it implies that the index of the pair contributing the largest fidelity-entropy loss for optimizing weights w_k primarily depends on the covariance (the denominator of equation 2.6). As a result, the pair indexed by $k \neq 1$ may have the highest-fidelity entropy, thereby allowing for the optimization of the associated weight coefficient w_k. Furthermore, in challenging tasks such as CIFAR-10, each training iteration yields an approximately equal number of correct and incorrect predictions. This balance causes the optimization contributions from individual samples to negate each other, resulting in nearly uniform weights after training.

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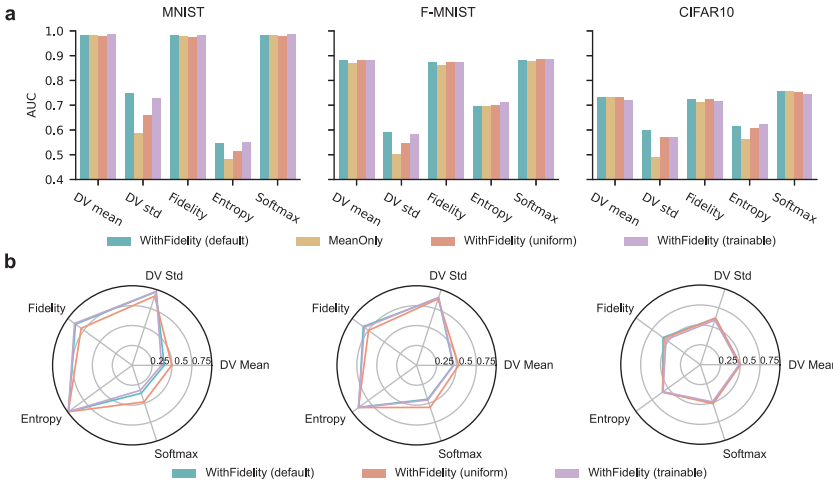


Figure 10: Comparison of the impact of various weight configurations on confidence metrics. (a) AUC that evaluates the models' capability to differentiate correct from incorrect predictions using different confidence metrics. (b) Proportion of samples in which a model can have an improved confidence expression using selected metrics after introducing the fidelity-entropy loss in training. Colors denote specific weight configurations.

We further assess the confidence metrics of these models as described in section 2.2. Figure 10a demonstrates that metrics based on the readout mean (DV Mean, Softmax, and Fidelity) show similar performance across models trained with different weight configurations, consistent with the conclusion in the main text. Compared to the MeanOnly model, those trained with the fidelity-entropy loss display a higher AUC using covariance-based metrics (DV Std and Entropy) for confidence, with the exception of Entropy in the F-MNIST data set. Given the specific weight configuration of WithFidelity, it unsurprisingly achieves a higher AUC with DV Std compared to Uniform and Trainable as it consistently sets a high priority in optimizing this quantity. By evaluating the proportion of input samples with which a model will achieve improved confidence measures following the incorporation of fidelity-entropy loss (see Figure 10b), we find that covariance-based metrics are optimized much better than the metrics that only involve the readout mean. Among different weight configurations, the improvement to confidence metrics is slightly less pronounced in the Uniform model as compared to the WithFidelity and Trainable models, while the difference between the latter two is minor.

B.2 Effect of Fidelity-Entropy Weight Configurations on Inference in SNN. We analyze the effects of different weight configurations in the

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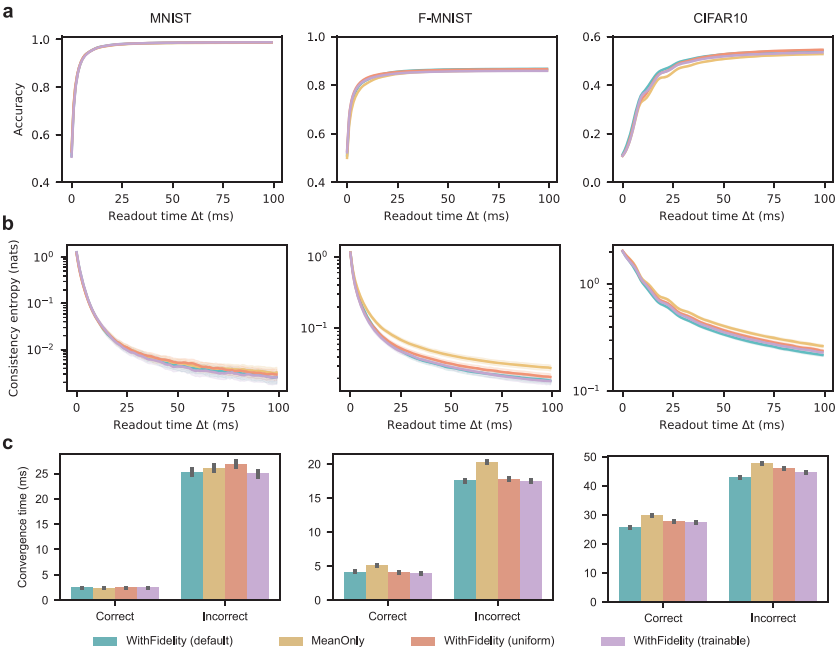


Figure 11: Impact of weight configurations on SNN dynamics during tasks. (a) Classification accuracy versus readout time. (b) Prediction consistency entropy versus readout time. (c) Average convergence time relative to the prediction results. The error bars indicate the 95% confidence intervals.

fidelity-entropy loss on the behavior of reconstructing SNN during inference. Using the same simulation settings as in section 2.3, we first assess the temporal dynamics of classification accuracy. As shown in Figure 11a, all models demonstrate similar classification performance in the limit of long readout time, but the MeanOnly model starts with a slightly slower growth in accuracy compared to models trained with fidelity-entropy loss, especially in difficult tasks (F-MNIST and CIFAR-10). For models utilizing various weight configurations in fidelity entropy, the default WithFidelity model demonstrates a marginally quicker convergence in accuracy compared to the other models on the CIFAR-10 data set. The superiority of the default weight configuration is further supported by prediction consistency (see Figure 11b). Generally the prediction consistency as measured by the entropy of prediction distribution for all models decreases with longer readout times, indicating a reduced trial-to-trial variability. Although the lower bound of this entropy is affected by the complexity of the task, the MeanOnly model exhibits an increased entropy (thus less consistency) in more challenging tasks compared to models trained with the

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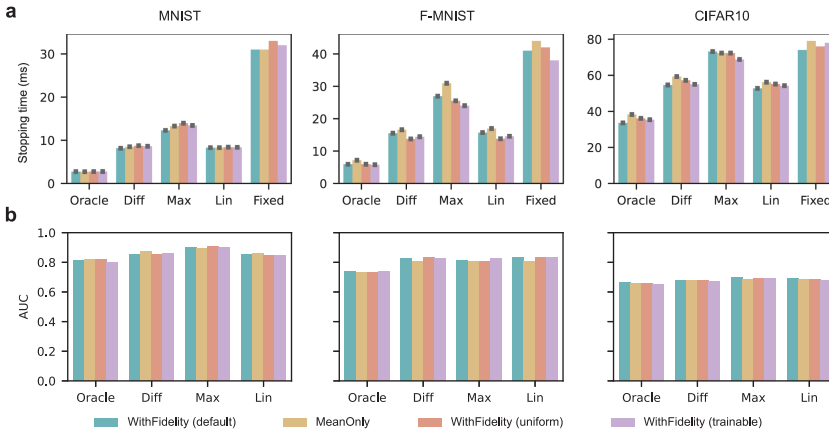


Figure 12: Comparison of results from applying different stopping policies on SNNs under various conditions. (a) Mean stopping time for each policy to achieve Oracle-level accuracy, with error bars representing the 95% confidence interval. (b) AUC scores of using stopping time to distinguish correct and incorrect predictions. The stopping time is obtained by stopping policies with the same decision thresholds used in panel b.

fidelity-entropy loss. Among various weight configurations of fidelity entropy, the WithFidelity model achieves a marginally better performance than the Uniform and Trainable models on the CIFAR-10 data set. By evaluating the average convergence time during inference, we find that correct predictions take consistently less time than incorrect predictions in various tasks and training conditions, as shown in Figure 11c. Incorporating fidelity-entropy loss generally reduces convergence times, especially in challenging tasks. Among the weight configurations, WithFidelity shows superior performance, particularly in the CIFAR-10 data set.

Next, we assess the efficacy of different stopping policies as applied to these models under different training conditions. As shown in Figure 12a, both the Difference and Linear policies can stop inference faster than the Max policy, achieving a classification performance comparable to the Oracle policy. Although the Max policy can also reduce the inference time, it is less effective on difficult tasks. Notably, these stopping policies are universally effective, regardless of the model's training approach. In addition, under higher task difficulty, models trained with the fidelity-entropy loss consistently exhibit shorter inference times than the MeanOnly model, independent of the stopping policy used. No single weight configuration for the fidelity-entropy loss is universally superior, each with advantages depending on tasks and policies. Furthermore, adaptive stopping policies reflect the models' decision confidence (see Figure 12b), as evidenced by all

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models' AUCs surpassing 0.5 across tasks, with no preference to specific training method.

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Data Availability

The data sets used in this study are available in the following databases. MNIST: https://git-disl.github.io/GTDLBench/datasets/mnist_datasets/; Fashion-MNIST: https://github.com/zalandoresearch/fashion-mnist; CIFAR-10 and CIFAR-100: https://www.cs.toronto.edu/~kriz/cifar.html; and ImageNet: https://www.image-net.org/download.php.

Code Availability

The code used in this article is available at https://github.com/BrainsoupFactory/moment-neural-network and https://github.com/ZhichaoZhu/moment-neural-network.

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